CariSurg MedTech Pathways - Week 0

Day 4: Clinical Write-Up

Vital Sign: Respiratory Rate (RR)

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Respiratory Rate (RR) is the number of breaths a person takes per minute and is measured by counting chest wall movements over sixty (60) seconds. It reflects the body's effort to maintain adequate gas exchange and acid-base balance, making it sensitive to disruptions across multiple organ systems, not just the lungs (Porter & Graham, 2025). The normal range for a resting adult is 12 to 20 breaths per minute (Porter & Graham, 2025). A rate below 12 breaths per minute is termed bradypnea and is commonly associated with opioid or sedative use, traumatic brain injury or alcohol intoxication, while a rate above 20 breaths per minute is termed tachypnea and may indicate pneumonia, sepsis, pulmonary embolism, metabolic acidosis or cardiac dysfunction (Porter & Graham, 2025). In triage, RR holds particular clinical weight because it is frequently the earliest objective indicator of patient deterioration, rising before blood pressure or oxygen saturation falls (Cretikos et al., 2008). It is one of the three Sequential Organ Failure Assessment (qSOFA) score, where a rate of 22 breaths per minute or more contributes to identifying patients at high risk of sepsis-related death or prolonged intensive care admission (Perman et al., 2020). Dispute its importance, RR remains the vital sign most prone to measurement error in emergency settings, with studies showing poor agreement between manual and automated recording (Lee et al., 2024).

References

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